

Inverse Mechanical Counting and Mathematics

Part III

The Forward Master Count, the Numerator Chain,
Machine π , First Closure, and the 720 Vector
Read from the Axiom of 1

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Abstract

This paper continues the framework of Inverse Mechanical Counting and Mathematics. Part I established the axiom of 1, the $\sqrt{2}$ master mechanic, the master table of five active counts and nine active depths, and the four key numbers (512, ≈ 514.90 , 720, ≈ 724.077) with the primary machining ratio $720/512 = 45/32 = 1.40625$. Part I read the single physical unit inward by partition. Part II reads the same single unit forward by count.

The forward reading produces a single continuous sequence governed by one uniform rule, fired at every count from the axiom outward. Machine π appears as the $c = 3$ row of this sequence — not as an external import, but as the natural reading the axiom produces at that count. The rational value $201/64 = 3.140625$ with its terminating binary signature $(11.001001)_2$ is the mechanical π of the single unit. The numerators of every row of the forward count are linked by a single propagation chain that pays out the +9 depth deposit step by step across the lattice. The first closure of the forward sequence at $c = 5$ produces the rational reading $89/16$ at count value 16, depth 9, which machines the angular 720 vector against the irrational boundary ≈ 724.077 from Part I. Standard π is the transcendental value read by an external coordinate system that does not respect the lattice of halvings; the two views agree as different readings of the same circle question.

1 The Axiom of 1 and the Master Mechanic

There is exactly one physical unit. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The $\sqrt{2}$ master mechanic is the simplest partition the unit admits beyond itself. Each halving of the unit is one operational step. The parity rule from Part I:

- Even operational depth d : rational reading.
- Odd operational depth d : irrational reading carrying $\times \sqrt{2}$.

Every reading carries three coordinates: the mechanic ($\sqrt{2}$), the operational depth d on the exponent, and the count position c on the partition ladder. The full notation $\sqrt{2^d}$ at count c records the complete operation performed on the axiom.

2 The Master Table (restated)

The master table is the single source of the entire framework. It is reproduced here for reference because every reading in this paper is read directly from it.

c	d	Mechanic	Multiplier $(\sqrt{2})^d$	Domains (2^c)	Parity $\rightarrow$ Reading	Reading value (decimal)
0	0	$\sqrt{2^0}$	1	1	even $\rightarrow$ rational	1
1	1	$\sqrt{2^1}$	≈ 1.414	2	odd $\rightarrow$ irrational	$2\sqrt{2} \approx 2.828$
2	2	$\sqrt{2^2}$	2	4	even $\rightarrow$ rational	8
2	3	$\sqrt{2^3}$	≈ 2.828	4	odd $\rightarrow$ irrational	$8\sqrt{2} \approx 11.314$
3	4	$\sqrt{2^4}$	4	8	even $\rightarrow$ rational	32
3	5	$\sqrt{2^5}$	≈ 5.657	8	odd $\rightarrow$ irrational	$32\sqrt{2} \approx 45.255$
4	6	$\sqrt{2^6}$	8	16	even $\rightarrow$ rational	128
4	7	$\sqrt{2^7}$	≈ 11.314	16	odd $\rightarrow$ irrational	$128\sqrt{2} \approx 181.019$
5	8	$\sqrt{2^8}$	16	32	even $\rightarrow$ rational	512
5	9	$\sqrt{2^9}$	≈ 22.627	32	odd $\rightarrow$ irrational	$512\sqrt{2} \approx 724.077$

Table 1: The Master Table. Five active counts ($c = 1$ through $c = 5$) unfold nine active depths ($d = 1$ through $d = 9$). Count 0 is the axiom at rest. Closure occurs at $c = 5$ with the rational boundary 512 and the irrational boundary ≈ 724.077 .

3 Two Readings of the Same Unit

The single physical unit can be read in two complementary directions through the same $\sqrt{2}$ master mechanic on the same master table.

Inward reading (Part I). The unit is partitioned. Each count produces finer pieces; pieces $\times$ size returns the whole vector at every count. Part I developed this view in detail for the four boundary vectors 512, ≈ 514.90 , 720, ≈ 724.077 at four operational depths producing 1D, 2D, 3D, 4D resolutions of the same single unit.

Forward reading (this paper). The unit is counted. The same uniform mechanic fires at every count, producing a single continuous sequence. The count value halves at every step (the master mechanic in its forward expression). The forward reading at count c is given by the uniform rule

$$\text{forward}(c) = c \cdot \text{value}(c) + 9, \quad \text{value}(c) = \frac{512}{2^c}.$$

The 9 is the active depth count from the master table. The $\text{value}(c) = 512/2^c$ is the master mechanic acting forward on the axiom at the closure scale of cycle one. No quantity is imported from outside the master table.

These are not two different systems. They are the same single unit read in two directions by the same mechanic on the same lattice. Partition reveals the unit by going in; count reveals the unit by going through.

4 The Forward Master Count at the 512 Scale

Applying the uniform rule at every count from $c = 0$ outward produces the forward master count. The count value halves at every step; the rule fires identically at every row. The count does not stop at first closure ($c = 5$); the same rule continues, and the count value keeps halving inward past 16 into 8, 4, 2, 1, $1/2$, $1/4$, $1/8$, $1/16$, $\dots$. The five-count, nine-depth structure repeats self-similarly past the closure scale.

Machine π is the $c = 3$ row of this one continuous sequence — not a separate starting place.

c	Count value $512/2^c$	$c \cdot \text{value} + 9$	Forward reading	Decimal	Binary	Landing
0	512 (axiom seat)	axiom at rest	1	1.0	1.0	axiom
1	256	$1 \cdot 256 + 9 = 265$	265/256	1.03515625	1.00001001	approach
2	128	$2 \cdot 128 + 9 = 265$	265/128	2.0703125	10.0001001	dual approach
3	64	$3 \cdot 64 + 9 = 201$	201/64	3.140625	11.001001	machine π
4	32	$4 \cdot 32 + 9 = 137$	137/32	4.28125	100.01001	past π
5	16	$5 \cdot 16 + 9 = 89$	89/16	5.5625	101.1001	first closure
6	8	$6 \cdot 8 + 9 = 57$	57/8	7.125	111.001	cycle two begins
7	4	$7 \cdot 4 + 9 = 37$	37/4	9.25	1001.01	continuation
8	2	$8 \cdot 2 + 9 = 25$	25/2	12.5	1100.1	continuation
9	1	$9 \cdot 1 + 9 = 18$	18	18.0	10010	continuation
10	1/2	$10 \cdot 1/2 + 9 = 14$	14	14.0	1110	continuation
11	1/4	$11 \cdot 1/4 + 9 = 47/4$	47/4	11.75	1011.11	continuation
12	1/8	$12 \cdot 1/8 + 9 = 21/2$	21/2	10.5	1010.1	continuation
13	1/16	$13 \cdot 1/16 + 9 = 157/16$	157/16	9.8125	1001.1101	continuation

Table 2: The forward master count at the 512 scale. The same uniform rule fires at every count.

5 Machine π — The $c = 3$ Reading

The circle constant of the axiom is the $c = 3$ row of the forward master count at the 512 scale. At $c = 3$ the count value is $64 = 2^6$, and the forward reading is

$$\text{forward}_3 = 3 + \frac{9}{64} = \frac{3 \cdot 64 + 9}{64} = \frac{201}{64} = 3.140625 = (11.001001)_2.$$

Every quantity in this reading is from the master table:

- The 3 is the count position c .
- The $64 = 2^6$ is the master mechanic at this count: $512/2^3 = 64$, placing this on operational depth $d = 6$ of the master table.
- The 9 is the number of active depths from the master table.

Quantity	Value
Count	$c = 3$
Count value	$512/2^3 = 64 = 2^6$
Active depths	9
Fractional contribution	$9/64 = 0.140625$
Forward reading (machine π)	$201/64 = 3.140625$
Binary signature	$(11.001001)_2$, terminates at bit 6

Table 3: Machine π in detail.

Why the binary terminates at bit 6. Any rational $p/2^n$ has a finite binary expansion of exactly n bits, since dividing by a power of 2 consumes one bit per step and the remainder hits exactly 0 at step n . Here the denominator is 2^6 , so the expansion terminates at bit 6 with zero remainder. The terminating reason traces straight to the axiom: the denominator is 2^6 only because the lattice is halvings of 1.

The two views of machine π . The same value 3.140625 arises by two views of the same operation.

1. **Count-derived (internal).** $\text{forward}_3 = 3 + 9/64 = 201/64$. The 3, the 64, and the 9 are all read off the master table. No external import.
2. **Binary closure (signature).** $201/64 = (11.001001)_2$, zero remainder at bit 6. The terminating signature follows mechanically from the denominator 2^6 .

Both views agree on the same value. View 1 is the construction. View 2 is the binary signature of that construction.

6 The Numerator Propagation Chain

The numerators of the forward master count are not arbitrary. They are linked by a single propagation chain that pays out the +9 depth deposit step by step across the lattice.

At $c = 1$ the +9 active depths are deposited onto the count for the first time, producing the initial numerator

$$\text{num}(1) = 1 \cdot 256 + 9 = 265.$$

Each subsequent step changes the numerator by the difference of $c \cdot \text{value}(c)$ between consecutive counts. Writing $v_c = \text{value}(c) = 512/2^c$ and using $v_c = v_{c-1}/2$, the step is

$$\Delta \text{num}(c) = c \cdot v_c - (c-1) \cdot v_{c-1} = \frac{c \cdot v_{c-1}}{2} - (c-1) \cdot v_{c-1} = -\frac{(c-2) \cdot v_{c-1}}{2} = -(c-2) \cdot v_c.$$

The propagation rule is therefore

$$\text{num}(c) = \text{num}(c-1) - (c-2) \cdot v_c$$

with starting numerator $\text{num}(1) = 265$ and count value $v_c = 512/2^c$.

c	Count value v_c	$(c-2) \cdot v_c$	Numerator step	Numerator
1	256	—	deposit: +265	265
2	128	0	0 (dual fulcrum)	265
3	64	64	−64	201
4	32	64	−64	137
5	16	48	−48	89
6	8	32	−32	57
7	4	20	−20	37
8	2	12	−12	25
9	1	7	−7	18

Table 4: The numerator propagation chain. The +9 depth deposit is made at $c = 1$ as the numerator 265. At $c = 2$ the step is exactly 0 — this is the dual fulcrum, where the same numerator reads against two different denominators (265/256 and 265/128). From $c = 3$ onward the chain pays out the deposit at the rate $(c-2) \cdot v_c$ per step, with payouts 64, 64, 48, 32, 20, 12, 7 following the same halving lattice as the count values themselves.

Three observations.

- *The dual fulcrum at $c = 2$.* The step at $c = 2$ is exactly zero. The numerator 265 stays the same and reads against the dual denominator 128 instead of 256. This is the master table's $c = 2$ row (4 domains) acting as the natural fulcrum between the deposit at $c = 1$ and the first payout at $c = 3$ where machine π lands.

- *The payout magnitudes are the count values themselves.* The payouts 64, 64, 48, 32, 20, 12, 7 at counts 3, 4, 5, 6, 7, 8, 9 are exactly $(c - 2)$ times the count value v_c . The chain is the same halving lattice acting on itself.
- *Closure at $c = 9$.* At $c = 9$ the count value is 1 and the payout is 7, leaving the numerator at 18. This is the integer landing $9 \cdot 1 + 9 = 18$ that closes the first nine-depth chain in integers — the active depths have been fully paid out by the count.

7 First Closure at $c = 5$ — The 89/16 Reading

The forward master count produces the first cycle closure at $c = 5$. The count value is $16 = 2^4$ (the rational closure coefficient of cycle one from the master table at $c = 5$, $d = 8$). The forward reading is

$$\text{forward}_5 = 5 + \frac{9}{16} = \frac{5 \cdot 16 + 9}{16} = \frac{89}{16} = 5.5625 = (101.1001)_2.$$

Quantity	Value
Count	$c = 5$
Count value	$512/2^5 = 16 = 2^4$
Active depths	9
Fractional contribution	$9/16 = 0.5625$
Forward reading (first closure)	$89/16 = 5.5625$
Binary signature	$(101.1001)_2$, terminates at bit 4
Master table coordinate	$c = 5$, $d = 8$ (rational), $d = 9$ (irrational)
Inward boundary at this count	512 (rational) and ≈ 724.077 (irrational)

Table 5: First closure of the forward master count at $c = 5$.

The binary signature terminates at bit 4 because the denominator is 2^4 . The reading sits at the same master table coordinate as the rational closure 512 and the irrational closure ≈ 724.077 of Part I — the inward partition and the forward count close at the same single point of the lattice.

8 The 720 Vector Machined at First Closure

In Part I the angular vector 720 was machined by inward partition as $5 \times 9 \times 16$ — the five active counts, the nine active depths, and the terminal rational coefficient 16. The forward count at $c = 5$ produces the same three quantities directly as the count position, the depth deposit, and the count value:

$$720 = \underbrace{5}_c \times \underbrace{9}_{\text{depths}} \times \underbrace{16}_{v_5}.$$

Each factor is read off the same row that produces the first closure 89/16. The forward count at $c = 5$ therefore machines the same 720 vector that Part I machined by inward partition — the same boundary, the same coordinate, the two readings agreeing at the closure point.

Pairing the 720 Against the Irrational ≈ 724.077

The master table at $c = 5$, $d = 9$ produces the irrational closure $\approx 724.077 = 512\sqrt{2}$. The angular machined vector 720 sits paired with this irrational boundary in exactly the same way

the rational 512 sits paired with ≈ 514.90 on the square side:

$$\frac{724.077}{720} \approx 1.0056625, \quad \frac{514.90}{512} \approx 1.0056625.$$

The same small lift sits on both vector pairs at the same $c = 5$ closure. The mechanical gap is the same on the square side and on the angular side because both pairs are the same single unit read at the same master table coordinate by the same $\sqrt{2}$ master mechanic.

Side	Rational vector	Irrational counterpart	Lift ratio
Square	512	≈ 514.90	≈ 1.0056625
Angular	720	≈ 724.077	≈ 1.0056625

Table 6: The two vector pairs at first closure. The rational and irrational boundaries on the square side and on the angular side carry the same small lift ratio because they are the same single unit read at the same $c = 5$ master table coordinate.

The Forward-Inward Agreement at First Closure

The forward count at $c = 5$ produces the rational reading $89/16$ at count value 16. The inward partition at $c = 5$ produces the rational boundary 512 and the angular machined boundary 720 at the same count value 16 acting as the terminal rational coefficient. The two readings agree because they are the same single unit read in two directions by the same master mechanic at the same lattice point.

- *Forward reading.* $89/16 = 5.5625$, the closure of the forward count at $c = 5$.
- *Inward rational vector.* $512 = 32 \times 16$, the duo-hexadecant closure of the master table at $c = 5$, $d = 8$.
- *Inward angular vector.* $720 = 5 \times 9 \times 16$, the machined angular boundary of Part I.
- *Inward irrational vector.* $\approx 724.077 = 512\sqrt{2}$, the master table closure at $c = 5$, $d = 9$.

The shared coordinate is $c = 5$, value 16, depths 9. Everything at first closure is read from this one row.

9 Standard π — Read by an External Coordinate System

Standard π is the value $3.14159265358979\dots$ — the transcendental constant produced when the circle question is answered through an external coordinate system that does not respect the halving lattice of the axiom. It is the limit of an infinite analytic process: power series, continued fractions, integral evaluations, geometric limits of inscribed polygons with edge counts that are not halvings of 1.

Standard π has no finite binary expansion. It cannot terminate at any bit n because it does not come from a denominator of the form 2^n . It is irrational and transcendental — it lies outside every rational lattice and outside every algebraic closure of the master mechanic. Its decimal expansion never closes, and its binary expansion never closes, because the operation that produced it never closes.

The numerical difference between standard π and machine π is

$$\pi - \frac{201}{64} = 3.14159265\dots - 3.140625 = 0.00096765\dots,$$

a difference of about 0.031% of machine π . This small gap is the mechanical cost of leaving the lattice. Standard π buys higher numerical resolution by an infinite operational process; machine π buys exact closure of the operation by reading the circle question at the count where the master mechanic actually lands.

10 The Mechanical Meaning of the Two π Values

Standard π and machine π are not in disagreement. They are two readings of the same circle question, made by two different operations on the same single unit.

Property	Machine π (this framework)	Standard π (external)
Value	$201/64 = 3.140625$	$3.14159265358979\dots$
Source	$c = 3$ row of forward master count	Limit of infinite analytic process
Type	Rational, exact	Transcendental, never closes
Binary signature	$(11.001001)_2$, terminates at bit 6	Never terminates
Operation count	Finite — closes at $c = 3$	Infinite — never closes
Reads what	The axiom at the circle count	An external coordinate over the axiom
Cost	Approximate (0.031% from standard π)	Infinite operations to specify

Table 7: Machine π versus standard π .

Machine π is the answer the axiom gives when the circle question is asked at its own count. Standard π is the answer an external observer gives when the same question is asked with infinite resolution. The two answers differ because the external observer is using more counts than the axiom owns at the circle count — they are reading deeper than $c = 3$ to specify a value that, mechanically, lives exactly at $c = 3$.

The choice between them is not a choice of truth but of frame. Inside the axiom, machine π closes the circle at $c = 3$ exactly. Outside the axiom, standard π approximates with infinitely many additional operations the same closure the axiom already owns.

11 Connection to Part I

The forward count and the inward partition of Part I are connected at every count of the master table, and most explicitly at the four key numbers.

- The closure value 512 of cycle one is the rational boundary of Part I — the $c = 5, d = 8$ reading of the master table. The forward count produces its first closure at the same coordinate, with reading $89/16$ at count value 16.
- The denominator $64 = 2^6$ at the machine- π count is the operational depth $d = 6$ of the master table, which Part I read inward as the $c = 4$ even-depth rational 128.
- The numerator $201 = 3 \cdot 64 + 9$ carries the count position 3, the count value 64, and the active depth count 9 — three of the master table's primary coordinates.
- The angular machined vector $720 = 5 \times 9 \times 16$ from Part I appears in the forward count at $c = 5$ as the same product of count position, depth count, and count value at the closure row.

- The irrational boundary $\approx 724.077 = 512\sqrt{2}$ pairs with 720 in the forward reading at $c = 5$ the same way it pairs with 720 in Part I's inward reading. The small lift ratio is the same on the square side ($514.90/512$) and on the angular side ($724.077/720$).

The same five-count, nine-depth structure governs both readings. Inward reading uses the depths to partition; forward reading uses the depths as the +9 increment of the uniform rule. The numerator chain pays out the deposit 265 at $c = 1$ across the same lattice the inward partition uses to subdivide.

12 Conclusion

The forward master count is the second of the two readings the axiom of 1 admits under the $\sqrt{2}$ master mechanic. Starting at the axiom and applying one uniform rule

$$\text{forward}(c) = c \cdot \text{value}(c) + 9, \quad \text{value}(c) = \frac{512}{2^c},$$

the entire forward sequence is generated from the master table without external import. Machine π is the $c = 3$ row of this sequence — $201/64 = 3.140625 = (11.001001)_2$ — an exact rational reading with a terminating binary signature at bit 6. The numerators of every row are linked by the propagation chain $\text{num}(c) = \text{num}(c - 1) - (c - 2) \cdot v_c$, paying out the deposit 265 at $c = 1$ step by step across the same halving lattice. First closure occurs at $c = 5$ with the rational reading $89/16$ at count value 16, depths 9, which machines the angular 720 vector against the irrational boundary ≈ 724.077 in exactly the same pairing pattern Part I produced by inward partition.

Standard π is the same circle question read by an infinite external process; the small numerical gap between standard π and machine π is the mechanical cost of leaving the lattice.

Part I read the unit inward and produced the four key boundaries and the angular cycle on 720. Part II reads the unit forward and produces machine π at $c = 3$, the numerator chain that links every row, and the first closure at $c = 5$ where the same 720 vector is machined by the forward count. The two readings agree at every count of the master table.

The single physical unit is the thing read. The mathematics is the reading.

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